

MAULANA AZAD NATIONAL INSTITUTE OF TECHNOLOGY BHOPAL
Department of Electrical Engineering

Mid-Term Exam (Oct. 2025)

Course: III Sem B.Tech

Sub: EMF& Materials Code: EE-24211

Time: 1 hour

MM: 20

Assume suitable data if necessary

Q. No.	Question	Marks	CO
Q1✓	Eight 25 nC point charges in free space are located symmetrically on a circle of radius 0.2m centered at origin in the $z = 0$ plane. (a) At what point on the z axis is $ E $ a maximum? (b) What is $ E _{\max}$	5	CO1
Q2	Derive the relation $\nabla \cdot \vec{D} = \rho_v$, where ' $\vec{D}$ ' is electric flux density and ρ_v is volume charge density.	5	CO2
Q3	Verify the 'divergence theorem for a field $\vec{f} = \frac{5r^2}{4} \hat{r}$ for spherical surface of radius 2m	5	CO1
Q4	If $\vec{E} = \left(\frac{-6y}{x^2} \right) \hat{x} + \left(\frac{6}{x} \right) \hat{y} + 5\hat{z}$, calculate V_{PQ} given $P(-7, 2, 1)\text{m}$ and $Q(4, 1, 2)\text{m}$	5	CO2